



# Dual boundary-spanning search and open innovation in China: The mediating role of organizational resilience and the moderating role of big data analytics capability

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## ABSTRACT

Open innovation is crucial for achieving high-quality and sustainable business growth. However, it remains unclear how prospective boundary-spanning search and responsive boundary-spanning search impact organizational resilience and open innovation in the VUCA (Volatility, Uncertainty, Complexity, and Ambiguity) new normal. Based on the resource-based theory and dynamic capabilities theory, this paper classifies boundary-spanning search into prospective boundary-spanning search and responsive boundary-spanning search to investigate the impact on open innovation and organizational resilience, and analyze the moderating role of big data analytics capability. The empirical study of 293 Chinese smart manufacturing firms indicates that both prospective and responsive boundary-spanning search have an inverted U-shaped impact on open innovation and organizational resilience. Organizational resilience partially mediates the relationships between prospective boundary-spanning search, responsive boundary-spanning search, and open innovation. Moreover, big data analytics capability moderates and flattens the inverted U-shaped relationships between prospective and responsive boundary-spanning search, and organizational resilience. This study explores the intrinsic mechanism of prospective and responsive boundary-spanning search on open innovation through organizational resilience. The findings enrich and expand the research on the antecedents of open innovation.

## 1. Introduction

In the context of the new normal characterized by Volatility, Uncertainty, Complexity, and Ambiguity (VUCA) (Martín-Rojas et al., 2023), firms face the imperative to adapt to dynamic environments and engage in continuous exploration and application of new knowledge through open innovation (Martín-Rojas et al., 2023). Open innovation refers to the firm's use of both internal and external complementary innovation resources to achieve innovation in the innovation process (Naqshbandi and Tabche, 2018; Sims et al., 2019). The essence of open innovation is that the firm breaks through the original organizational boundaries, conducts extensive and in-depth scanning and searching of the external environment in which it is located, and emphasizes the integration and utilization of both internal and external innovation resources of the firm. Compared to closed innovation, open innovation

emphasizes collaboration and sharing, aiming to break down traditional barriers and obstacles in innovation models (Chesbrough, 2003; West and Gallagher, 2006). External innovation forces can be brought in to accelerate the pace and enhance efficiency. Exposure to a wider range of innovation sources can also increase the diversity and possibilities of internal innovation (West et al., 2014; West and Gallagher, 2006). Simultaneously, open innovation assists firms in adapting to market changes by maintaining close contact with consumers, suppliers, and other stakeholders (Hung and Chou, 2013; Lopez-Vega et al., 2016; Wang et al., 2023; West and Bogers, 2014). This enables firms to quickly learn about market needs and trends, allowing them to adjust the direction of innovation and ensure that their products meet market demands. Thus, firms engaged in open innovation cannot rely solely on their existing knowledge to meet the demands of innovation; but must also tap into external knowledge resources (Garriga et al., 2013; Qu

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et al., 2023).

Resource-based theory posits that unique and valuable knowledge is a crucial resource for firms to gain a competitive edge (Chadwick et al., 2015; Zhu et al., 2023). To infuse their operations with innovative elements, firms need to capture heterogeneous knowledge keenly (Sirmon et al., 2011). Boundary-spanning search represents a knowledge-seeking activity that firms undertake across existing organizational or technological boundaries (Yang et al., 2021; Zhu et al., 2023). Boundary-spanning search is an important means for firms to acquire heterogeneous knowledge and overcome internal rigidities and capability traps, facilitating the establishment of innovation advantages (Zhu et al., 2023). Proactive boundary-spanning search represents a leading method for anticipating and preparing for future changes and trends. In contrast, a reactive, passive search, also known as a responsive boundary-spanning search, is designed to gather and analyze information when changes in the external environment occur or when events have already had an impact (Yang et al., 2021). Previous studies have analyzed the significant role of external knowledge acquisition in innovation (Cammarano et al., 2019; Houessou et al., 2023; Wang et al., 2020). However, few have explored the relationship between boundary-spanning search, specifically prospective and responsive boundary-spanning search, and open innovation in terms of search timing. Therefore, investigating the impact of prospective and responsive boundary-spanning search on firms' open innovation is a worthwhile endeavor.

Furthermore, the survival and prosperity of an organization depends heavily on its ability to adapt to a dynamic environment (Dubey et al., 2023). The resource-based theory places considerable emphasis on the role of resources in fostering innovation (Martín-Rojas et al., 2023). However, it lacks a clear articulation of the specific mechanisms through which resources drive innovation (Teece et al., 1997). As an extension of resource-based theory, dynamic capabilities theory indicates that firms' innovations are contingent upon not only the resources themselves, but also their dynamic capabilities (Martín-Rojas et al., 2023). The core of dynamic capabilities theory is to focus on how firms can adapt and optimize resources allocation to achieve innovation in a changing environment (Teece et al., 1997). Dynamic capabilities theory suggests that in a turbulent external environment, if the resources that previously provided the firm with a competitive advantage prove difficult to adapt to changing environmental demands, it will lead to core rigidity and impede the sustainable development of the firm (Dubey et al., 2023; Khlystova et al., 2022; Napier et al., 2024). Organizations face continuous shocks from the external environment during their operations, requiring organizational resilience to effectively manage complex and evolving contingencies (Lin and Fan, 2024). Organizational resilience, as a dynamic capability, enables firms to efficiently and flexibly manage external knowledge, anticipate potential threats, and respond effectively to risks that may arise in the process of open innovation (Martín-Rojas et al., 2023). Previous research on the factors influencing organizational resilience has mainly focused on leadership characteristics (Tuazon et al., 2021), organizational resources (Do et al., 2022). Organizational resilience relies heavily on external knowledge, and a wealth of external knowledge enables organizations to quickly identify and respond to potential crises, facilitating their rapid recovery, rebound, and even reversal (Dan et al., 2017; Napier et al., 2024). Consequently, it is necessary to investigate the mechanism of prospective and responsive boundary-spanning search on organizational resilience. In addition, organizational resilience, as the dynamic ability of organizations to withstand, recover and grow from external shocks, also plays a key role in fostering open innovation (Hillmann and Guenther, 2021). Therefore, it is imperative to elucidate the influence of prospective and responsive boundary-spanning search on organizational resilience and explore its mediating role between prospective and responsive boundary-spanning search and open innovation.

In today's highly competitive and uncertain business environment, the dynamic capabilities theory provides a powerful framework for

understanding organizational resilience. Dynamic capabilities refer to a firm's ability to sense, capture, and allocate internal and external resources to respond to rapidly changing environments (Teece et al., 1997). Meanwhile, prospective boundary-spanning search and responsive boundary-spanning search are strategies for firms to achieve organizational resilience. Prospective boundary-spanning search enables firms to proactively cross organizational boundaries to explore new technologies and market trends in a wider range of areas (Dan et al., 2017). This proactive exploration allows firms to proactively perceive opportunities and threats that may affect their development (Lopez-Vega et al., 2016). For example, BYD Auto foresaw the rise of the new-energy vehicle market through prospective boundary-spanning search. It established close cooperation with battery suppliers, developed battery R&D and production plans in advance, and attracted relevant technical talents to join the firm. This integration of internal and external resources effectively enhanced its competitiveness in the new-energy vehicles market and its ability to cope with market changes. Responsive boundary-spanning search enables firms to pay close attention to the dynamics of competitors and industry leaders and access timely their experiences of success and failure in technological innovation, market strategy, etc. (Yang et al., 2021). When a firm discovers that a competitor launches a new product or adopts a new business model, it can expedite, learn from, and draw on its useful elements and avoid taking a detour in its own development process. For instance, Xiaomi's drew on Apple's industrial design concepts and Samsung's supply chain management model. It launched cost-effective mobile phone models like the Redmi series and made localized improvements to meet the demands of China's sinking market. By real-time monitoring the product iteration speed of competitors such as Huawei and OPPO, Xiaomi shortened the industry's average 18-month R&D cycle to 9 months and rapidly launched benchmark-setting models. Through responsive boundary-spanning search strategy, Xiaomi witnessed a remarkable growth in its mobile phone market share in India, surpassing Samsung to become the top-selling brand.

The tenets of dynamic capabilities theory posit that firms engage in the construction, integration, and allocation of internal and external resources in a dynamic environment with the objective of coping with changes in the external environment (Ferreira et al., 2020; Napier et al., 2024; Teece et al., 1997). In an evolving external market environment and dynamic internal organizational process, the firms dynamic capability to effectively respond to changes in the internal and external environments and resource consumption is the key factor determining its competitive advantage (Napier et al., 2024). In the context of digital economy, big data has emerged as a strategic resource and valuable information asset that enhances open innovation capability of firms (El-Kassar and Singh, 2019; Gupta and George, 2016; Shamim et al., 2020). Effective resource management is essential for the survival and development of organizations operating in dynamic environments (Eisenhardt and Martin, 2000; Mikalef et al., 2019). While resource management provides a robust foundation for organizational operations, the accelerated evolution of information technology has precipitated a correspondingly rapid industrial iteration and the emergence of a vast quantity of data (Huynh et al., 2023). This has rendered traditional resource management techniques insufficient for navigating the intricate and dynamically evolving market landscape (Zhang et al., 2021). The value of big data presents a significant window of opportunity, and the ability of firms to rapidly identify, analyze, and process big data becomes a powerful adjunct to traditional resource management (Wang et al., 2020). Big data analytics capability is used to describe an organization's ability to effectively collect, store, process, analyze, and extract valuable information from large, diverse, complex, and rapidly generated data (Huynh et al., 2023). This capability is an essential component of dynamic capability that encompasses not only the mastery of data technology but also a profound comprehension of data, utilization of analytical instruments, generation of insights, and the capacity to make data-driven decisions (Wang et al., 2020). Big data analytics

capability enables firms to effectively analyze and interpret large-scale, complex, and high-dimensional data sets (AlNuaimi et al., 2021; Huynh et al., 2023). As the foundation for other digital technologies, big data analytics capability can provide organizations with more immediate insights and actionable references for decision-making. The functionality of technologies such as artificial intelligence (AI), machine learning, and blockchain is contingent upon the availability of high-quality, large-scale data inputs and data analysis outputs. Big data analytics provides data support for other technologies and is the foundation of intelligent decision-making processes (Arias-Perez et al., 2021). This serves as a valuable tool to harness credible resources from big data and support decision-making processes. By leveraging big data analytics capability, firms can facilitate seamless transmission and exchange of data across various levels (Huynh et al., 2023). This enables information and knowledge acquired through prospective and responsive boundary-spanning search to be stored, integrated, and processed in a cost-effective and efficient manner, thus optimizing resource allocation (Gupta and George, 2016; Mikalef et al., 2019). Big data analytics capability not only provide technical support for boundary-spanning search but also enhance the speed and quality of firms' processing and application of acquired knowledge. Existing research suggested that the use of big data analytics capability can improve the efficiency and effectiveness of boundary-spanning search (Xue and Wang, 2024). Consequently, organizations with higher capabilities in big data analytics have greater potential to transform external knowledge resources obtained through boundary-spanning search, fostering organizational resilience (Appiah, 2024; Xue and Wang, 2024). Thus, it is necessary to investigate the role of big data analytics capability and uncover their moderating effect on the relationship between prospective and responsive boundary-spanning search and organizational resilience.

This paper employs resource-based theory and dynamic capabilities theory to examine how prospective and responsive boundary-spanning search enhances organizational resilience and promotes open innovation. This study aims to highlight the importance of organizational resilience and big data analytics capability in fostering open innovation, while also considering the mediating role of the former and the moderating role of the latter. To achieve this goal, this study conducted empirical research in China's smart manufacturing firms using a questionnaire survey. By uncovering the underlying mechanism through which prospective and responsive boundary-spanning search influences firm open innovation, this study contributes to the existing research on the antecedents of open innovation. Moreover, this research provides a strong theoretical foundation for firms to engage in open innovation.

The remainder of the paper is structured as follows: Section 2 presents a theoretical model that offers a plausible account of the relationship between prospective and responsive boundary-spanning search, organizational resilience, big data analytics capability, and open innovation. Section 3 delineates the methodology employed to evaluate the hypotheses. Section 4 presents the results of the analysis. Section 5 presents the discussions, theoretical implications, managerial implications and future research.

## 2. Theoretical background and hypotheses

### 2.1. Boundary-spanning search and open innovation

The main idea behind open innovation is to efficiently combine internal and external innovation resources of firms, and accelerate deployment of technologies to market through collaborative partnerships and market channels (Hung and Chou, 2013; Wang et al., 2023; West and Bogers, 2014). Boundary-spanning search, which involves acquiring knowledge across organizational boundaries, is crucial for open innovation (Dan et al., 2017; Ogink et al., 2023). Boundary-spanning search can be categorized into two types: prospective and responsive boundary-spanning search (Yang et al., 2021; Zhu et al., 2023).

Prospective boundary-spanning search enables firms to stay ahead of their competitors and actively explore knowledge resources that have not yet been widely accepted in the market but have potential value (Dan et al., 2017; Lopez-Vega et al., 2016). As previously observed by Singh and Fleming (2010) and McDonald and Eisenhardt (2019), the prioritization of access to resources has the potential to enhance the diversity and novelty of a firm's resource pool, thereby affording firms greater flexibility to innovate. Through prospective boundary-spanning search, firms can take the lead in realizing technological advancement and obtaining a first-mover advantage (Garriga et al., 2013). Therefore, prospective boundary-spanning search not only helps firms to improve learning efficiency and shorten the learning curve, but also breaks through their own resources and technological bottlenecks to explore the potential needs of users and future market opportunities (Chesbrough and Brunswicker, 2014; Katila and Chen, 2008). More importantly, prospective boundary-spanning search reduces the likelihood of competitors taking a first-mover advantage and provides strong support for firms to establish unique competitive advantages. However, it is important to note that excessive prospective boundary-spanning search may have a negative impact on open innovation (Chunhsien et al., 2020). The hidden nature, complexity, and uncertainty of external knowledge make search difficult and may incur opportunity costs, i.e. distract firms (Dahlander et al., 2016; Khadijeh et al., 2020). For prospective boundary-spanning search, excess search may result in knowledge that is too complex and novel to be integrated with existing knowledge, leading to blocked innovation (Flor et al., 2018; Wang et al., 2023). Given that firms have limited attention spans, excessive complexity can result in redundant knowledge within a firm, leading to wasted resources and hindering the conversion of new knowledge into innovation value. Therefore, we propose the following hypotheses.

**H1a.** The impact of prospective boundary-spanning search on open innovation follows an inverted U-shaped curve.

In contrast, responsive boundary-spanning search enables firms to follow their competitors and search for knowledge resources already available in the market. As Jeong and Kim's (2019) research indicates, responsive boundary-spanning search that adhere to established technological trajectories are conducive to enhancing the legitimacy of firms, facilitating their access to external heterogeneous resources in a more seamless manner and enabling them to innovate. Thus, responsive boundary-spanning search helps firms respond to market changes in a timely manner and reduces innovation risks. Through responsive boundary-spanning search, firms can obtain knowledge that is more in line with their own knowledge base, form new knowledge portfolios, and promote open innovation (Yang et al., 2021). At the same time, it can help firms identify knowledge to improve existing products or processes and increase customer recognition and acceptance, thereby facilitating open innovation. As responsive boundary-spanning search deepens, firms will encounter challenges associated with cognitive rigidity and resources homogenization. In this way, firms will overly rely on established cognitive patterns and ways of thinking, as well as competitors' practices, while underestimating the distinctive qualities and potential of innovations. This may impede the flow of open innovation. Additionally, excessive responsive boundary-spanning search can lead to information overload and increased costs, thus negatively impacting open innovation. Moreover, excessive external information can hinder rational resource allocation of firms, leading to neglect of key knowledge integration and application, and missing opportunities for innovation. Therefore, we propose the following hypotheses.

**H1b.** The impact of responsive boundary-spanning search on open innovation follows an inverted U-shaped curve.

### 2.2. Boundary-spanning search and organizational resilience

Organizational resilience refers to an organization's ability to

quickly adapt, recover, and even surpass its original state of development in response to internal and external disturbances and threats (Parker and Ameen, 2018). It has become a crucial factor in a firm's stable development in a complex and ever-changing business environment. Dynamic capabilities theory suggests that firms can develop new capabilities by effectively acquiring and integrating internal and external resources to better adapt to environmental changes and achieve their development goals (Xue and Wang, 2024). Therefore, in a business environment full of uncertainty, firms can enhance organizational resilience by responding effectively to various challenges.

Prospective boundary-spanning search is a proactive activity that encourages firms to cross organization boundaries and access heterogeneous information or resources that are not yet widely accepted (Katila and Chen, 2008; Yang et al., 2021). Accessing to valuable resources is a necessary but not a sufficient condition for firms to establish an innovative advantage. Dynamic capabilities theory underscores the significance of firms integrating and leveraging resources. On the one hand, prospective boundary-spanning search can broaden a firm's development thinking and effectively alleviates internal bottlenecks and strategic crises, and facilitate firms to increase their knowledge breadth and anticipate external shocks. In this way, firms can formulate corresponding coping strategies (Dan et al., 2017). On the other hand, prospective boundary-spanning search encourages firms to establish a close cooperation network with universities and other leading research institutions. This helps to update existing knowledge base of the firm and enhances organizational resilience, making it more capable of coping with external shocks and breaking organizational rigidity (Tasic et al., 2020). However, it is important to note that when boundary-spanning search exceeds a certain level, firms may face problems such as information overload and increased difficulty in knowledge integration (Dahlander et al., 2016). These problems can lead to less efficient organizational decision-making and slower response times, which in turn weakens organizational resilience (Wang et al., 2023). In the case of prospective boundary-spanning search, lack of experience may result in unnecessary resource development waste and increased potential risks and failures due to excessive search activities. Additionally, overly complex external networks may consume too much time and effort, thus affecting firms' ability to adapt and recover quickly.

**H2a.** The effect of prospective boundary-spanning search on organizational resilience follows an inverted U-shaped curve.

At the same time, responsive boundary-spanning search aims to deepen firms' understanding and utilization of their existing knowledge base (Yang et al., 2021). Research has shown that increased depth of knowledge significantly improves organizational resilience, enabling firms to better cope with challenges (Wang et al., 2023). Responsive boundary-spanning search enables firms to learn from their partners' successes in dealing with past external shocks, to strengthen internal cross-functional communication and collaboration based on those experiences, enables firms to break down vertical and horizontal boundaries, and to strengthen organizational resilience. This way enables firms to reduce operating costs and strengthens their ability to respond (Parker and Ameen, 2018). However, firms that rely too heavily on responsive boundary-spanning search and neglect prospective boundary-spanning search may struggle to adapt to the complex and ever-changing external environment (Ferreras-Méndez et al., 2016). Over-reliance on responsive boundary-spanning search can result in missed opportunities for proactive adjustment and optimization, and increase the knowledge gap. Additionally, excessive pursuit of responsive boundary-spanning search tends to limit the scope of firms, resulting in the acquisition of largely homogenized resources that lack uniqueness and heterogeneity. The redundancy of resources and the paucity of heterogeneous resources can easily lead to cognitive myopia. In other words, excessive responsive boundary-spanning search can lead to high levels of knowledge homogenization, which may increase the cost of knowledge integration, reduce cooperation efficiency, and

generate redundant knowledge that could decrease firms' sensitivity to external threats. Therefore, we propose the following hypotheses.

**H2b.** The effect of responsive boundary-spanning search on organizational resilience follows an inverted U-shaped curve.

### 2.3. Organizational resilience and open innovation

The core of dynamic capabilities theory is that firms not only need to have a good resource base, but also need to have the ability to quickly perceive opportunities and threats, seize opportunities, and reconfigure resources in a timely manner (Schilke, 2014; Teece et al., 1997). Organizational resilience refers to the ability of an organization to quickly adapt, recover, and rebound in response to internal and external shocks or changes (Hung and Chou, 2013; Ogink et al., 2023). This ability enables firms to maintain flexibility and stability in uncertain environments while also capitalize on new opportunities through learning and innovative development (Kantur and Iseri-Say, 2015).

In uncertain environments, external shocks often manifest as random and unpredictable changes (Chakrabarti, 2015). These shocks may originate from sudden crisis, market fluctuation, policy changes, economic crisis or technological disruption, among others (Li et al., 2025). Shocks in uncertain environments place higher demands on organizational resilience, as firms need to be flexible in adjusting their resources and coping strategies in unknown directions (Napier et al., 2024). Table 1 shows the differentiation framework for organizational resilience.

Dynamic capabilities theory emphasizes the continuous learning and adaptability of firms (Eisenhardt and Martin, 2000; Schilke, 2014). Open innovation is characterized by active collaboration with external partners to bring in new ideas and resources, as opposed to the traditional internal innovation process. This form of innovation emphasizes

**Table 1**  
Differentiation framework for organizational resilience.

| Type of shock             | Speed of response required                                                                                                                                                                 | Core competences                                                                                                                                                                  |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sudden Crisis             | Need for rapid response, short decision-making time frames with immediate action, e.g. firms' swift adjustment of the office mode in the office mode, etc.                                 | Flexibility, resource deployment, communication and coordination, e.g. rapid establishment of telecommuting mode, supplier coordination, etc.                                     |
| Market Fluctuation        | Need for accelerated response, timely response to competitors' actions, e.g. traditional automotive firms need to adjust their strategies rapidly to Tesla's impact.                       | Innovation ability, market insight, branding ability, e.g. traditional automotive firms to increase R&D investment, emphasizing the brand advantage, etc.                         |
| Policy Changes            | Proactive foresight to evolving policy dynamics and adaption to the policy adjustments. For instance, align R & D strategies with new energy.                                              | Policy interpretation ability, strategic adjustment ability, resource integration ability, e.g. monitoring policies, seeking policy support, adjusting the direction of R&D, etc. |
| Economic crisis           | Balancing within a set timeframe short-term cost control and long-term market expansion, e.g. economic recession, firms need to control costs in a timely manner, open up new markets.     | Cost control ability, market expansion ability, financial risk management ability, such as streamlining personnel, expanding overseas markets, optimizing debt structure, etc.    |
| Technological disruptions | Rapid adaptation and pre-emptive planning, in the emergence of new technologies can be quickly followed up or transformation, e.g. mobile manufacturers response to smartphone innovation. | Technology R&D capability, strategic transformation capability, sensitivity to new technologies, e.g. advance layout of R&D, rapid launch of innovative products, etc.            |



the integration of external knowledge into an organization's internal resources, breaking down barriers to progress. Firstly, organizational resilience enables firms to maintain stable operations and quickly adapt to changes in the face of uncertainty and challenges, while effectively absorbing new technologies and knowledge from external sources, thus enhancing their sensitivity to external innovation opportunities (Garrido-Moreno et al., 2024; Liang and Li, 2024). This is how organizational resilience enables firms to more effectively identify potential collaboration opportunities in volatile market environments, thereby promoting the realization of open innovation. Secondly, organizational resilience emphasizes the adaptability and flexibility of firms, which can not only enhance their ability to accommodate external innovation resources, but also actively establish connections with diverse external partners (Liang and Li, 2024). This accelerates the transformation of open innovation achievements and improves the efficiency of the firm's innovation processes. In addition, organizational resilience is manifested in a firm's capacity to expeditiously recuperate in the event of a crisis (Garrido-Moreno et al., 2024; Liang and Li, 2024; Shela et al., 2024). The capacity to withstand risks such as technological failure or market non-acceptance is a key benefit of organizational resilience (Liang and Li, 2024; Shela et al., 2024). It enables firms to better cope with these challenges and maintain a positive attitude towards external cooperation and innovation in the face of failure, thereby ensuring the continued advancement of open innovation. Organizational resilience enables firms to integrate newly acquired external knowledge with internal resources, and breaks down barriers of traditional innovation models, thus facilitating open innovation (West and Bogers, 2014). In summary, organizational resilience is a key enabler of open innovation by fostering adaptability, enhancing resource integration, encouraging collaboration, and ensuring continuity in the face of challenges. Therefore, we propose the following hypotheses.

**H3.** The effect of organizational resilience on open innovation is positive.

#### 2.4. The mediating role of organizational resilience

Dynamic capabilities theory posits that simply acquiring external resources is insufficient for firms to innovate effectively. Adapting to changes in the external environment is also a necessary condition for innovation. YahiaMarzouk and Jin (2022) highlighted the importance of resource acquisition in the innovation process of firms, as well as the difficulty of achieving organizational strategic goals if these resources are not effectively utilized. Meanwhile, the dynamic capabilities theory posits that the perceived capability, integration capability, and coordination capability of a firm are crucial elements that comprise its dynamic capabilities. Organizational resilience can be defined as the capacity of the firm to perceive and adapt to environmental changes, coordinate internal and external resources to withstand external shocks, and learn from lessons to achieve rebound and recovery. Accordingly, organizational resilience represents a crucial element of the firm's dynamic capabilities and a key driver for achieving open innovation. We thus argue that organizational resilience mediates the relationship between boundary-spanning search and open innovation, as appropriate boundary-spanning search facilitates the formation of organizational resilience, which is a key component of open innovation.

Prospective boundary-spanning search is a crucial way for firms to access a diverse knowledge base by interacting across boundaries. This contributes to accessing heterogeneous resources (Lopez-Vega et al., 2016), which in turn compensates for the lack of internal innovation resources and enhances the firm's ability to cope with market changes and foster organizational resilience (Wang et al., 2023). Prospective boundary-spanning search not only is beneficial for firms to search tacit, irreplaceable, and potentially valuable knowledge (Yang et al., 2021), but also can help firms obtain more innovation resources, reduce the risk of knowledge inertia, and enhance the adaptability of firms to the

external environment. Accordingly, firms can quickly adjust their strategies and processes in the face of external turbulence and threats, enhancing organizational resilience (Napier et al., 2024). This resilience ensures the firm's ability to remain stable and sustainable in the face of external shocks (Lin and Fan, 2024; Xue and Wang, 2024). By combining search opportunities with open innovation strategies, firms can achieve their innovation goals. Furthermore, prospective boundary-spanning search can be more effective in identifying external market and environmental changes, allowing firms to access information ahead of competitors (Wang et al., 2023). This increased environmental sensitivity and predictive power can enhance organizational resilience and facilitate the identification of potential collaborators for obtaining resources, information, and support to jointly advance the innovation process.

Responsive boundary-spanning search focuses on finding complementary knowledge resources that match the original knowledge base and are readily available, deepening the rational allocation and utilization of existing knowledge by the firm (Yang et al., 2021). Organizational resilience ensures that firms maintain stable operations by adapting to industry trends or competing with rivals, leveraging accumulated knowledge resources, managing redundant information, improving the efficiency of knowledge and resource utilization, and ultimately fostering open innovation. In this way, organizational resilience enables firms to identify warning signs, re-evaluate past successes and failures of stakeholders (Hillmann and Guenther, 2021; Lin and Fan, 2024), utilize partner experiences to adjust product solutions and processes in a timely manner, and achieve long-term development goals through open innovation. Therefore, we propose the following hypotheses.

**H4a.** Organizational resilience serves as a mediator in the relationship between prospective boundary-spanning search and open innovation.

**H4b.** Organizational resilience serves as a mediator in the relationship between responsive boundary-spanning search and open innovation.

#### 2.5. The moderating role of big data analytics capability

Dynamic capabilities theory emphasizes the need for firms to possess the ability to adapt and learn dynamically in response to environmental changes, including insight, prediction, and feedback (Teece et al., 1997). Insight refers to a firm's capacity to integrate internal and external data to deepen its understanding of market trends, customer needs, and the potential of internal resources (Eisenhardt and Martin, 2000). Prediction involves firms using big data analytics to build predictive models, identify potential opportunities and risks, and formulate response strategies in advance (Gupta and George, 2016). Feedback means continuously monitoring market dynamics and making timely strategy adjustments through big data analytics to enhance the firm's adaptability speed and flexibility. From this, it can be seen that dynamic capabilities theory regards big data as a valuable organizational resource. The capability to analyze big data is an organizational capability based on data analytics. It enables firms to analyze generic knowledge from a vast volume of big data and transforms it into specific knowledge resources required for firms' innovation. In contrast to general dynamic capabilities, big data analytics capability aims to enhance an organization's competitive advantage by providing new insights. This is not an ad-hoc practice but rather a continuous process of adaptation and change. Organizations can acquire external heterogeneous resources, which help them anticipate change and proactively adapt to uncertainty. Big data analytics capability refers to the exclusive insights derived from capturing, storing, and analyzing data through the efficient utilization of technology and talent within an organization (Ferraris et al., 2019; Huynh et al., 2023; Mikalef et al., 2019). In the digital economy era, big data analytics capability is not only a matter of technical proficiency, but also encompasses the capability to manage vast data assets and synchronize operational endeavors (AlNuaimi et al.,

2021). Consequently, we posit that organizational resilience is not solely influenced by boundary-spanning search, but also significantly depends on a firm's big data analytics capability. Additionally, big data analytics capability serves as a moderating factor in the inverted U-shaped relationship between boundary-spanning search and organizational resilience.

Specifically, when firms possess stronger big data analytics capability, they are able to more precisely identify potential prospective boundary-spanning search opportunities through deep data mining and meticulous interpretation of vast amounts of information. This capability enhances firms to effectively integrate and utilize external knowledge resources (Cadden et al., 2023; Huynh et al., 2023; Sivarajah et al., 2024), thereby shortening learning curve, mitigating external uncertainties, and fostering organizational resilience. Furthermore, stronger big data analytics capability allows firms to monitor and accurately predict the risks associated with prospective boundary-spanning search activities in real-time. As a result, firms are better able to assess the risk level and modify appropriate response strategies for prospective boundary-spanning search, averting undue risks and maintaining risk levels within manageable boundaries. Consequently, the negative impact of excessive prospective boundary-spanning search on organizational resilience is mitigated. Conversely, when firms have a low level of big data analytics capability, they may struggle to effectively distill and integrate new knowledge gained through prospective boundary-spanning search. This can lead to slower knowledge acquisition, poor knowledge internalization, and a higher risk of knowledge misapplication, ultimately affecting organizational adaptability and undermining resilience (Cadden et al., 2023; Rialti et al., 2019). Additionally, without robust data support, cross-sector collaboration and decision-making processes may become overly reliant on subjective judgments and past experiences, leading to inefficient coordination and challenges in managing cultural conflicts (AlNuaimi et al., 2021; Gupta and George, 2016). This can increase internal chaos and further weaken organizational resilience. Therefore, a low level of big data analytics capability not only diminishes the positive impact of prospective boundary-spanning search on organizational resilience but also exacerbates the negative consequences of excessive prospective boundary-spanning search.

With a high level of big data analytics capability, firms are able to capture market changes in real-time, enabling them to swiftly formulate (Mikalef et al., 2019; Sivarajah et al., 2024), and adjust strategies in response to responsive boundary-spanning search. This confers a high level of responsiveness and adaptability, which significantly enhances organizational resilience. Furthermore, firms with stronger big data analytics capability can precisely identify boundary-spanning targets, thus avoid blind trends and efficiently allocating resources. Accordingly, the negative consequences of resource fragmentation on organizational resilience will be prevented. In essence, a high level of big data analytics capability amplifies the positive impact of responsive boundary-spanning search on organizational resilience and mitigates the negative effects of excessive responsive boundary-spanning search. Instead, a low level of big data analytics capability can lead to slow sensing of market changes and competitive dynamics (Cadden et al., 2023; Lin and Fan, 2024; Rialti et al., 2019). This slows down firms' response speed, making firms prone to imprecise responsive boundary-spanning search, potentially missing critical timing or responding inappropriately. Additionally, inadequate big data analytics capability restricts firms' ability to efficiently extract knowledge from vast heterogeneous sources. This inefficient knowledge management in further undermines the application of sensing new knowledge and business renewal, thereby weakening organizational resilience. Consequently, low level of big data analytics capability not only reduces the positive impact of responsive boundary-spanning search on organizational resilience but also exacerbates the negative consequences of excessive responsive boundary-spanning search. Therefore, we propose the following hypotheses.

**H5a.** Big data analytics capability serves as a moderator in the relationship between prospective boundary-spanning search and organizational resilience.

**H5b.** Big data analytics capability serves as a moderator in the relationship between responsive boundary-spanning search and organizational resilience. Fig. 1 shows theoretical model of our research.

### 3. Methods

#### 3.1. Data and samples

We utilized a questionnaire to collect data. To establish the target audience for our questionnaire, we referenced the classification of the smart manufacturing industry outlined by the National Bureau of Statistics within their Classification of China's Strategic Emerging Industries. Smart manufacturing, an industry heavily reliant on knowledge, maintains close communication and cooperation with external stakeholders. Therefore, our survey targets encompassed several subdomains within smart manufacturing, including new energy vehicles, high-end equipment manufacturing, new materials, artificial intelligence, and biopharmaceuticals. China is a major global manufacturing center with rapid growth and large - scale smart manufacturing industry. It boasts abundant firm cases and data resources, which provide an adequate sample base for our study. Smart manufacturing firms are highly reliant on technological innovation and cross-border cooperation. In the face of technological changes and market competition, they frequently engage in boundary-spanning search activities, which makes the study results more typical and representative. As significant proponents of data-driven transformation and innovation in China, smart manufacturing firms achieve comprehensive intelligence in production, management, and services through utilizing advanced technologies, including modern information technology and intelligent equipment. Smart manufacturing companies are typically situated at the vanguard of technological advancement, confronting rapidly evolving market conditions and technological developments. In order to maintain their competitive advantage, it is necessary for these firms to conduct constant boundary-spanning searches to identify new market opportunities, potential threats, and competitive dynamics. In light of the above situation, the investigation into the activities of smart manufacturing firms offers insights into the function of prospective and responsive boundary-spanning search in the context of open innovation. Furthermore, smart manufacturing firms possess a substantial quantity of data resources, which can be leveraged to achieve comprehensive monitoring and optimization of the production process through big data analytics. The richness and analyzability of such data provide invaluable resources for studying firms' open innovation.

In order to obtain representative sample data, this paper combines the distribution characteristics of smart manufacturing firms and the focus of China's regional economic development. The sample is comprised of middle and senior managers from firms in the key regions of Beijing, Shanghai, Zhengzhou, Guangdong, Chongqing, and Xi'an. Specifically, Beijing is the capital of China, with a relatively complete industrial system and a high level of economic development. Shanghai is one of the pioneering demonstration zones of China's reform and opening-up, and also one of the most economically developed and innovative regions in China. Zhengzhou is one of the densely populated regions in China, with relatively balanced economic development, a certain industrial base and human resources. Guangzhou is one of the most developed and dynamic regions in the Chinese economy, with a certain level of industrial development and human resources. Guangzhou is one of the most economically developed and dynamic regions in China, with a well-developed industrial system and an advanced manufacturing industry. Chongqing is an important strategic support point for the development of China's western region, with a solid

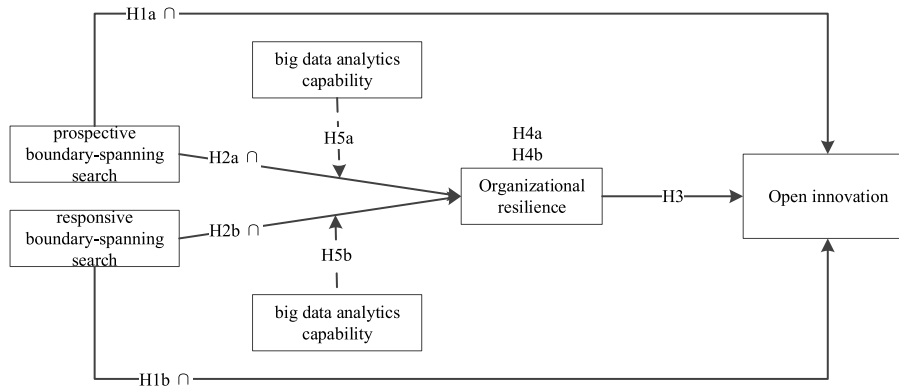


Fig. 1. Theoretical model.

industrial foundation, and is an important base for electronic equipment manufacturing and high-end equipment manufacturing in China. Xi'an is an important strategic depth and new growth point of China's economic development, with a nationally important equipment manufacturing base and a scientific and technological innovation center. Each of these cities plays an important role in China's economic development, and offers valuable case study for examining the varying degrees of open innovation and prospective and responsive boundary-spanning search strategies employed by smart manufacturing firms.

The survey respondents were primarily middle and senior managers from firms located in key regions such as Beijing, Shanghai, Zhengzhou, Guangdong, Chongqing, and Xi'an. Given that the questionnaire delves into the strategic development of these firms, we prioritized respondents who possess a clear understanding of their organizations' current status and strategic plans.

In terms of questionnaire design, we referenced and translated scales from foreign literature, ensuring the accuracy of our expressions. Additionally, we invited two experts in the field of innovation to review the questionnaire, identifying and refining any unclear or irrelevant items. Their suggestions also guided us in adjusting the item order. Prior to its official distribution, we conducted a pilot survey among 20 MBA students from our university, using the feedback to further refine the questionnaire.

We distributed the formal questionnaire using both offline and online methods. Firstly, between October and December 2023, we distributed paper questionnaires to middle and senior managers of firms enrolled in MBA and EMBA programs at various institutions. A total of 100 questionnaires were distributed, and 97 were successfully recovered. Secondly, leveraging our team's extensive social network, we emailed electronic questionnaires to firms from November 2023 to January 2024. Of the 494 questionnaires distributed, 275 were returned.

In total, we distributed 594 questionnaires, and 372 were returned, resulting in an actual recovery rate of 62.6 %. After rigorous screening to exclude missing values, multiple-choice answers, and randomly filled invalid questionnaires, we ultimately counted 293 valid questionnaires, representing an effective recovery rate of 78.8 %. The detailed characteristics of the sample firms, covering aspects such as firm size, firm age, region, industry type, and the ownership structure, are presented in Table 2.

### 3.2. Variables and measurement

To ensure the sample's representativeness, this study used existing scales to measure variables. Prospective boundary-spanning search and responsive boundary-spanning search were measured by utilized Katila and Chen (2008); Yang et al. (2021)'s four-item scale. Prospective boundary-spanning search includes "our firm often identifies new opportunities in emerging markets before its competitors", "our firm

Table 2  
Basic information of sample firms.

| Firm characteristics | Classification                      | Frequency | Percentage |
|----------------------|-------------------------------------|-----------|------------|
| Firm size (sales)    | 10 million and below                | 44        | 15.02 %    |
|                      | 10 million-30 million               | 113       | 38.57 %    |
|                      | 30 million-100 million              | 55        | 18.77 %    |
|                      | 100 million-200 million             | 60        | 20.48 %    |
|                      | 200 million and above               | 21        | 7.16 %     |
| Firm age (years)     | 5 years and below                   | 17        | 5.80 %     |
|                      | 6–10 years                          | 52        | 17.75 %    |
|                      | 11–15 years                         | 64        | 21.84 %    |
|                      | 16–20 years                         | 91        | 31.06 %    |
|                      | 21 years and above                  | 69        | 23.55 %    |
| Geographical region  | Beijing                             | 72        | 24.57 %    |
|                      | Shanghai                            | 55        | 18.77 %    |
|                      | Zhengzhou                           | 43        | 14.68 %    |
|                      | Guangdong                           | 27        | 9.22 %     |
|                      | Chongqing                           | 31        | 10.58 %    |
| Industry type        | Xi'an                               | 65        | 22.18 %    |
|                      | New Energy Vehicles                 | 43        | 14.68 %    |
|                      | High-end Equipment Manufacturing    | 54        | 18.43 %    |
|                      | New Materials                       | 72        | 24.57 %    |
|                      | Artificial Intelligence             | 88        | 30.03 %    |
| Ownership structure  | Biopharmaceuticals                  | 36        | 12.29 %    |
|                      | State-owned firms                   | 42        | 14.33 %    |
|                      | Private firms                       | 178       | 60.75 %    |
|                      | Foreign-funded firms                | 21        | 7.17 %     |
|                      | Chinese-foreign joint venture firms | 52        | 17.75 %    |

frequently develops innovative ways to meet customer needs ahead of our competitors", "our firm often introduces product features that are unique and not offered by our competitors", "we often pioneer new production processes that our competitors lack". Responsive boundary-spanning search includes "our firm often follows competitors in enhancing the competitiveness of existing products in the market", "we often emulate competitors to improve the quality of our existing products", "our firm frequently adopts strategies followed by competitors to enhance the production efficiency of our existing products", "we often emulate competitors to refine the development process of our existing products".

Organizational resilience is measured by incorporated Kantur and Iseri-Say (2015)'s six-item scale, items are "our firm consistently succeeds in devising diverse solutions to problems", "our firm always demonstrates a relentless drive towards continuous growth", "our firm possesses the capability to sustain this female shipping operation effectively", "our firm is proficient in capitalizing on market opportunities to fuel our growth", "our firm is more adept at acquiring resources to bolster its growth momentum", "our firm is capable of enhancing

access to resources to drive growth”.

For big data analytics capability, we employed Gupta and George (2016); Mikalef et al. (2019) 's scale, comprising eight items, “our firm has access to vast amounts of unstructured and rapidly evolving data”, “we are able to integrate data from diverse sources into a unified database for seamless access”, “our firm explores and utilizes cloud services for data processing and analytical purposes”, “our firm possesses exceptional expertise in managing structured data”, “our firm’s managers are proficient in understanding and applying insights derived from big data”, “our big data analysts and other departments regularly engage in cross-departmental meetings to synchronize their efforts”, “our employees have undergone training on big data”, “our workforce effectively extracts meaningful insights from web-based data”.

Open innovation is designed by referring Hung and Chou (2013)’s study, encompassed four items, “our firm maintains a close exchange of information with external sources of innovation”, “our firm obtains a significant portion of its technology from external sources”, “our firm tends to share technology with external parties”, “our firm strives to establish stronger ties with external parties and relies on them for innovation”.

To enhance the reliability of our findings, we included industry type, firm sales, and firm age as control variables. Firstly, considering the crucial impact of industry type on innovation, our sample encompassed five diverse industries: new energy vehicles, high-end equipment manufacturing, new materials, artificial intelligence, and biopharmaceuticals. Consequently, we controlled for industry type by coding four dummy variables, with new energy vehicles-related industries serving as the reference group. Secondly, since firm age shapes strategic goals, we accounted for this variable, measuring it based on the year of establishment. Finally, given that firm sales influence innovation investment, we controlled for this factor, quantifying it through annual sales figures. All items, except for the control variables, were measured using a 7-point Likert scale, ranging from 1 (“not at all”) to 7 (“completely”).

### 3.3. Common variance bias

We employed three methods to assess the potential impact of common variance biases (Hair et al., 2011; Podsakoff and Organ, 1986). Firstly, we measured each variable through multiple question items in our questionnaire design, ensuring that antecedent and outcome variables were placed in distinct sections. Secondly, we conducted a Harman’s one-factor test to determine the extent of common variance bias. The results revealed that the first factor accounted for 39.787 % of the total variance, falling below the 50 % threshold. This indicates that the issue of common variance bias in our study is not significant. Thirdly, we tested the fit of the one-factor model using confirmatory factor analysis. The results demonstrated poor model fit. This further corroborates our findings and confirms that there are no serious common variance bias issues in our sample data.

### 3.4. Reliability and validity

Reliability test is typically evaluated by using two key coefficients: the composite reliability (CR) coefficient and the internal consistency reliability coefficient (Cronbach’s  $\alpha$  coefficient). Prior to calculating the CR coefficients and Cronbach’s  $\alpha$  for prospective boundary-spanning search, responsive boundary-spanning search, organizational resilience, big data analytics capability, and open innovation, confirmatory factor analysis was conducted for each research variable. The results of the model fitting for the research variables are presented in Table 3. As shown in Table 3, the fit indices of the measurement models for the research variables reached an acceptable range, indicating that the measurement models of the variables align well with the sample data.

We conducted an exploratory factor analysis of the data using SPSS 21.0. First, the KMO (Kaiser - Meyer - Olkin) measure and Bartlett’s test

**Table 3**

Results of fitting the measurement model for the study variables.

|                                      | $\chi^2/df$ | GFI   | NFI   | TLI   | CFI   | RMSEA |
|--------------------------------------|-------------|-------|-------|-------|-------|-------|
| Prospective Boundary-spanning Search | 2.144       | 0.975 | 0.978 | 0.979 | 0.983 | 0.002 |
| Responsive Boundary-spanning Search  | 2.175       | 0.967 | 0.971 | 0.972 | 0.975 | 0.046 |
| Organizational Resilience            | 2.313       | 0.993 | 0.998 | 0.998 | 0.999 | 0.031 |
| Big data analytics capability        | 2.145       | 0.946 | 0.969 | 0.970 | 0.973 | 0.064 |
| Open Innovation                      | 1.153       | 0.983 | 0.987 | 0.990 | 0.991 | 0.017 |

of sphericity were performed on the questionnaire data to determine whether the study variables were suitable for exploratory factor analysis. As shown in Table 4, the KMO value of this study was greater than 0.8, and the result of Bartlett’s test of sphericity was significant. This indicates that the questionnaire data is suitable for exploratory factor analysis. Second, principal component analysis was employed to extract the common factors with eigenvalues greater than 1. Subsequently, the varimax rotation method was used to rotate these common factors to obtain the loadings of each measurement item on the common factors. As can be seen from Table 4, factor loadings are greater than 0.5, and the number of items for each common factor is greater than 2. These items effectively reflect the information of the common factors.

With regard to the reliability test, Table 4 illustrates that the CR coefficients of prospective boundary-spanning search, responsive boundary-spanning search, organizational resilience, big data analytic capabilities, and open innovation were all greater than 0.9, which is greater than the threshold of 0.7. It can therefore be concluded that the measurement scales of each research variable demonstrate good composite reliability. The Cronbach’s  $\alpha$  coefficient of each research variable exceeded the threshold of 0.7, thereby indicating that the measurement scales of the research variables exhibit good internal consistency. The measurement scales of prospective boundary-spanning search, responsive boundary-spanning search, organizational resilience, big data analytics capability and open innovation are reliable.

Three validities tests were conducted in this study. First, confirmatory factor analysis was used to test construct validity. The Kaiser-Meyer-Olkin (KMO) values for the variables exceeded 0.800, and Bartlett’s test yielded a significance of 0.000, confirming their suitability for factor analysis. Second, the average variance extraction (AVE) surpassed 0.700, indicating strong convergent validity in the questionnaire and an ideal model quality. Third, confirmatory factor analysis was performed to assess discriminant validity. The results yielded a reasonable fit. As displayed in Table 5, compared the five-factor model with other factor model, the five-factor model was the best ( $\chi^2/df = 2.096$ , IFI = 0.956, TLI = 0.951, CFI = 0.956, RMSEA = 0.064), indicating a satisfactory discriminant validity.

## 4. Analysis and results

### 4.1. Descriptive statistics analysis

Table 6 displays the means, standard deviations, and correlation coefficients of the variables. Notably, all correlation coefficients among the variables are below 0.5, with variance inflation factor (VIF) values well below the upper limit of 10 and tolerances exceeding the lower limit of 0.100. This indicates the absence of multicollinearity among the variables. The observed correlations lay the foundation for subsequent hypotheses testing.

### 4.2. Direct effect test

We employed hierarchical regression analysis to examine our hypotheses and assess the inverted U-shaped relationship, as advocated by



**Table 4**  
Results of composite reliability and validity analysis.

| Variables                                  | Items | Factor loadings | Cronbach's $\alpha$ | KMO   | CR    | AVE   |
|--------------------------------------------|-------|-----------------|---------------------|-------|-------|-------|
| Prospective Boundary-spanning Search (PBS) | PBS1  | 0.923           | 0.943               | 0.872 | 0.959 | 0.854 |
|                                            | PBS2  | 0.921           |                     |       |       |       |
|                                            | PBS3  | 0.926           |                     |       |       |       |
|                                            | PBS4  | 0.926           |                     |       |       |       |
| Responsive Boundary-spanning Search (RBS)  | RBS1  | 0.853           | 0.891               | 0.842 | 0.929 | 0.765 |
|                                            | RBS2  | 0.881           |                     |       |       |       |
|                                            | RBS3  | 0.879           |                     |       |       |       |
|                                            | RBS4  | 0.885           |                     |       |       |       |
| Organizational Resilience (OR)             | OR1   | 0.892           | 0.947               | 0.939 | 0.931 | 0.810 |
|                                            | OR2   | 0.897           |                     |       |       |       |
|                                            | OR3   | 0.906           |                     |       |       |       |
|                                            | OR4   | 0.889           |                     |       |       |       |
|                                            | OR5   | 0.909           |                     |       |       |       |
|                                            | OR6   | 0.907           |                     |       |       |       |
| Big data analytics capabilities (BDAC)     | BDAC1 | 0.907           | 0.972               | 0.843 | 0.976 | 0.835 |
|                                            | BDAC2 | 0.889           |                     |       |       |       |
|                                            | BDAC3 | 0.917           |                     |       |       |       |
|                                            | BDAC4 | 0.914           |                     |       |       |       |
|                                            | BDAC5 | 0.918           |                     |       |       |       |
|                                            | BDAC6 | 0.922           |                     |       |       |       |
|                                            | BDAC7 | 0.916           |                     |       |       |       |
|                                            | BDAC8 | 0.926           |                     |       |       |       |
| Open Innovation (OI)                       | OI1   | 0.934           | 0.934               | 0.861 | 0.953 | 0.835 |
|                                            | OI2   | 0.926           |                     |       |       |       |
|                                            | OI3   | 0.889           |                     |       |       |       |
|                                            | OI4   | 0.906           |                     |       |       |       |

**Table 5**  
Discriminant validity.

| Model              | Variable                   | $\chi^2/df$ | IFI   | TLI   | CFI   | RMSEA |
|--------------------|----------------------------|-------------|-------|-------|-------|-------|
| One-factor model   | PBS + RBS + RO + BDAC + OI | 14.961      | 0.437 | 0.425 | 0.471 | 0.219 |
| Three-factor model | PBS + RBS, RO + BDAC, OI   | 8.437       | 0.721 | 0.694 | 0.720 | 0.160 |
| Four-factor model  | PBS + RBS, RO, BDAC, OI    | 4.204       | 0.881 | 0.868 | 0.881 | 0.105 |
| Five-factor model  | PBS, RBS, RO, BDAC, OI     | 2.098       | 0.956 | 0.951 | 0.956 | 0.064 |

Haans et al. (2016). To establish this relationship, we set  $Y = \beta_0 + \beta_1 X + \beta_2 X^2$ . The inverted U-shaped relationship must satisfy three prerequisites: first,  $\beta_2$  must be negative and statistically significant; second, the slope of the curve  $\beta_1 + 2\beta_2 X$  must be positive when the independent variable X attains its minimum value, and negative when it reaches its maximum value; and third, the turning point of the curve, calculated as  $-\beta_1/2\beta_2$ , must fall within the range of value observed for the independent variable.

To test hypotheses H1a and H1b, we have designated open innovation as the dependent variable. In Model 3, the coefficient of PBS is positive, while the coefficient of its quadratic term is negative and statistically significant ( $\beta = -0.190$ ,  $p < 0.01$ ). Upon centering the data, the

range of PBS values becomes  $[-2.510, 3.240]$ . When PBS attains its minimum value, the slope is 1.315; conversely, at its maximum value, the slope is  $-0.870$ . Notably, the curve's turning points of 0.950 falls within this range, indicating a significant inverted U-shaped impact on open innovation. This finding supports hypothesis H1a.

Similarly, the coefficient of RBS is positive, with its quadratic term also negative and statistically significant ( $\beta = -0.154$ ,  $p < 0.01$ ). Regarding RBS, the centered value range is  $[-2.970, 2.530]$ . When RBS is at its minimum, the slope is 0.909, and at its maximum, the slope is  $-0.785$ . The curve's turning point,  $-0.019$ , lies within this range, indicating a significant inverted U-shaped influence on open innovation. This supports hypothesis H1b.

To test hypotheses H2a and H2b, we designated organizational resilience as the dependent variable. In Model 8, the coefficient of PBS is positive, while the coefficient of its quadratic term is negative and statistically significant ( $\beta = -0.520$ ,  $p < 0.001$ ). When PBS attains its minimum value, the slope is 2.96; whereas at its maximum value, the slope is  $-3.020$ . The turning points of the curve, 0.336, falls within the range of values for PBS. This suggests that PBS has a significant inverted U-shaped impact on organizational resilience, thereby supporting hypothesis H2a.

The coefficient of RBS is positive, and its quadratic term coefficient is negative and significant ( $\beta = -0.105$ ,  $p < 0.05$ ). When RBS reaches its minimum value, the slope is 0.607; at its maximum value, the slope is  $-0.548$ . The turning points for RBS,  $-0.030$ , lies within its value range.

**Table 6**  
Means, stand variances and correlations.

| Variables   | 1         | 2        | 3        | 4            | 5            | 6            | 7            | 8            |
|-------------|-----------|----------|----------|--------------|--------------|--------------|--------------|--------------|
| 1. Industry | <b>1</b>  |          |          |              |              |              |              |              |
| 2. Sales    | -0.046    | <b>1</b> |          |              |              |              |              |              |
| 3. Age      | -0.066    | 0.068    | <b>1</b> |              |              |              |              |              |
| 4. PBS      | -0.617*** | -0.069*  | 0.137**  | <b>0.924</b> |              |              |              |              |
| 5. RBS      | -0.361*** | -0.020   | 0.021    | 0.355***     | <b>0.875</b> |              |              |              |
| 6. OR       | -0.077    | -0.017   | 0.092    | 0.271***     | 0.168**      | <b>0.900</b> |              |              |
| 7. BDAC     | 0.122**   | 0.020    | -0.062   | -0.267***    | -0.185**     | -0.583***    | <b>0.914</b> |              |
| 8. OI       | -0.093    | -0.136** | 0.084    | 0.290***     | 0.174**      | 0.323***     | -0.272***    | <b>0.967</b> |
| MEANS       | 3.430     | 2.860    | 3.320    | 3.762        | 4.224        | 3.558        | 3.944        | 3.901        |
| S.D.        | 1.334     | 1.423    | 1.519    | 1.865        | 1.327        | 1.625        | 1.811        | 1.843        |

Note:  $n = 293$ , \* $p < 0.05$ , \*\* $p < 0.01$ , \*\*\* $p < 0.001$ ; diagonally bolded data are AVE mean root values.

Additionally, the turning points of  $-0.081$  also falls within the range of values for RBS. This indicates that RBS has a significant inverted U-shaped effect on organizational resilience, supporting hypothesis H2b.

To test hypotheses H3, we designated open innovation as the dependent variable. In Model 4, the coefficient of OR is positive and statistically significant ( $\beta = 0.239$ ,  $p < 0.001$ ). This finding supports hypothesis H3.

#### 4.3. Mediating effects test

To examine the mediating role of organizational resilience in the relationship between PBS, RBS, and open innovation, we drew upon the mediating effect testing methodology pioneered by Baron and Kenny (1986). Model 5 demonstrates that organizational resilience has a statistically significant positive impact on open innovation ( $\beta = 0.138$ ,  $p < 0.05$ ). When comparing Model 3 with Model 5, we observe that the absolute value of the quadratic term coefficient for PBS has decreased and remains significant ( $\beta = -0.119$ ,  $p < 0.05$ ), while the absolute value of the quadratic term coefficient for RBS decreases and maintains significance ( $\beta = -0.139$ ,  $p < 0.05$ ). These findings suggest that the influence of both PBS and RBS on open innovation diminishes after accounting for organizational resilience. In essence, organizational resilience partially mediates the relationship between PBS, RBS and open innovation, thereby supporting hypotheses H4a and H4b.

Furthermore, this study utilized the Medcurve macro program to delve deeper into the nonlinear mediating role of organizational resilience in the relationship between PBS, RBS and open innovation. After conducting 5000 repeated samples with 95 % confidence intervals, the findings are summarized in Table 8. Specifically, the instantaneous mediating effects of organizational resilience at varying levels of PBS—low (M-1SD), medium (M), and high (M+1SD)—are 0.296, 0.061, and  $-0.174$ , respectively. Notably, the 95 % confidence intervals consistently exclude zero, firmly establishing the nonlinear mediating role of organizational resilience between PBS and open innovation. This finding reinforces hypothesis H4a.

Similarly, when examining the mediating effects of organizational resilience at different levels of RBS—low, medium, and high—we observe instantaneous mediating effects of 0.175, 0.035, and  $-0.106$ , respectively. While the mediating effect is supported by the 95 % confidence intervals excluding zero at both low and high levels of RBS, it is not supported at the medium level, as the confidence interval includes zero. This result partially verifies the nonlinear mediating role of organizational resilience in the relationship between RBS and open innovation, further validating hypothesis H4b.

#### 4.4. Moderating effects test

To investigate the moderating influence of BDAC on the inverted U-shaped relationship, we employed the research approach outlined by Haans et al. (2016). The moderation of the inverted U-shaped manifests in two scenarios. Firstly, in the curvilinear relationship  $Y = \beta_0 + \beta_1X + \beta_2X^2 + \beta_3XZ + \beta_4X^2Z + \beta_5Z$ , the direction of the curve's shift—whether to the right or left—is determined by the sign of  $\beta_1\beta_4 - \beta_2\beta_3$ . Secondly, the steepness or flatness of the curve is judged based on the positivity or negativity of  $\beta_4$ .

As evident from the results presented in Model 10 and Model 11 in Table 7, BDAC significantly moderate the inverted U-shaped relationship between PBS ( $\beta = 0.394$ ,  $p < 0.001$ ) and RBS ( $\beta = 0.114$ ,  $p < 0.05$ ) with regard to organizational resilience. Specifically, this moderation flattens the inverted U-shaped relationship. Furthermore, our calculations reveal that as BDAC increase, the apex of organizational resilience shifts to the right (0.059), whereas the apex of organizational resilience shifts to the left (0.003). These findings validate hypotheses H5a and H5b (see Fig. 2).

To illustrate the moderating effect of BDAC on the relationship between PBS, RBS and organizational resilience more vividly, we plotted

**Table 7**

Instantaneous mediating effect test.

|           | XVAL  | 95 % confidence interval |          | SE    | THETA    |
|-----------|-------|--------------------------|----------|-------|----------|
|           |       | Boot LCI                 | Boot UCI |       |          |
| PBS→OR→OI | 1.897 | 0.027                    | 0.569    | 0.138 | 0.296    |
|           | 3.762 | 0.004                    | 0.121    | 0.029 | 0.061    |
|           | 5.627 | $-0.344$                 | $-0.020$ | 0.083 | $-0.174$ |
| RBS→OR→OI | 2.849 | 0.089                    | 0.288    | 0.051 | 0.175    |
|           | 4.224 | $-0.005$                 | 0.087    | 0.024 | 0.035    |
|           | 5.598 | $-0.236$                 | $-0.013$ | 0.057 | $-0.106$ |

the trend of the impact of these variables under varying levels of BDAC. Specifically, we plotted the trends based on the mean of BDAC plus or minus one standard deviation, as depicted in Figs. 2 and 3.

#### 4.5. Robustness test

The CLC Estimator tool is designed to estimate latent constructs using congeneric approaches, providing more rigorous measurement of latent variables compared to simpler methods like taking averages or sums (Mcneish and Wolf, 2020). Therefore, we recalculated the latent variables and tested the robustness of the model using the CLC estimator tool proposed by Marzi et al. (2023). In CLC Estimator tool, we chose EFA maximum likelihood estimation as our model estimation method and weighted average as the score estimation method. The results of the confidence test are shown in Table 9. The results obtained using CLC Estimator tool are basically consistent with the previous results.

We used the CLC Estimator tool to calculate the latent variables and conducted robustness tests. The findings were found to be generally consistent with the previous results, with direct, mediating and moderating effects being tested. The test results are shown in Table 10.

In Model III, the coefficient of PBS is positive, while the coefficient of its quadratic term is negative and statistically significant ( $\beta = -0.190$ ,  $p < 0.01$ ). After centering the data, the range of PBS values becomes  $[-2.512, 3.238]$ . When PBS attains its minimum value, the slope is 1.316; conversely, at its maximum value, the slope is  $-0.869$ . Notably, the curve's turning points of 0.950 falls within this range, indicating that PBS has a significant inverted U-shaped impact on open innovation. This finding further supports hypothesis H1a.

Similarly, the coefficient of RBS is positive, with its quadratic term also negative and statistically significant ( $\beta = -0.155$ ,  $p < 0.01$ ). Regarding RBS, the centered value range is  $[-2.512, 3.238]$ . When RBS is at its minimum, the slope is 0.917, and at its maximum, the slope is  $-0.791$ . The curve's turning point,  $-0.016$ , lies within this range, indicating that RBS has a significant inverted U-shaped influence on open innovation. This finding further supports hypothesis H1b.

In Model VIII, the coefficient of PBS is positive, while the coefficient of its quadratic term is negative and statistically significant ( $\beta = -0.519$ ,  $p < 0.001$ ). When PBS attains its minimum value, the slope is 1.149; whereas at its maximum value, the slope is  $-0.680$ . The turning points of the curve, 1.100, falls within the range of values for PBS. This finding further supports hypothesis H2a.

On the other hand, the coefficient of RBS is positive, and its quadratic term coefficient is negative and significant ( $\beta = -0.105$ ,  $p < 0.05$ ). When RBS reaches its minimum value, the slope is 0.607; at its maximum value, the slope is  $-0.549$ . The turning points for RBS,  $-0.080$ , lies within its value range. Additionally, the turning points of  $-0.081$  also falls within the range of values for RBS. This finding further supports hypothesis H2b.

In Model IV, the coefficient of OR is positive and statistically significant ( $\beta = 0.239$ ,  $p < 0.001$ ). This finding further supports hypothesis H3.

When comparing Model III with Model V, we observe that the absolute value of the quadratic term coefficient for PBS decreases and remains significant ( $\beta = -0.118$ ,  $p < 0.05$ ), while the absolute value of

**Table 8**  
Regression results of hierarchical model.

|                              | OI       |          |          |          |          | OR       |          |           |           |           |           |
|------------------------------|----------|----------|----------|----------|----------|----------|----------|-----------|-----------|-----------|-----------|
|                              | Model 1  | Model 2  | Model 3  | Model 4  | Model 5  | Model 6  | Model 7  | Model 8   | Model 9   | Model 10  | Model 11  |
| <b>Control Variable</b>      |          |          |          |          |          |          |          |           |           |           |           |
| Industry 2                   | −0.101   | 0.073    | −0.040   | −0.128   | −0.035   | 0.109    | 0.227*** | −0.032    | −0.015    | −0.065    | −0.008    |
| Industry 3                   | 0.014    | 0.049    | −0.037   | −0.055   | −0.057   | 0.291*** | 0.315*** | 0.138     | 0.090     | 0.098     | 0.098     |
| Industry 4                   | 0.139    | 0.309**  | 0.087    | 0.060    | 0.094    | 0.330*** | 0.443*** | −0.056    | −0.016    | −0.071    | −0.003    |
| Industry 5                   | −0.244** | 0.106    | −0.002   | −0.233** | 0.005    | −0.046   | 0.192    | −0.025    | −0.014    | −0.054    | 0.006     |
| Firm sales                   | −0.172** | −0.133*  | −0.125*  | −0.161** | −0.124*  | −0.044   | −0.018   | −0.005    | −0.006    | −0.022    | −0.003    |
| Firm age                     | 0.055    | 0.035    | 0.030    | 0.044    | 0.026    | 0.045    | 0.032    | 0.032     | 0.024     | 0.048     | 0.024     |
| <b>Independent variables</b> |          |          |          |          |          |          |          |           |           |           |           |
| PBS                          |          | 0.283*** | 0.361*** |          | 0.313*** |          | 0.182**  | 0.350***  | 0.236***  | 0.243***  | 0.229**   |
| RBS                          |          | 0.067*   | −0.006   |          | −0.004   |          | 0.070*   | −0.017    | −0.037    | −0.046    | −0.012    |
| <b>Squared</b>               |          |          |          |          |          |          |          |           |           |           |           |
| PBS <sup>2</sup>             |          |          | −0.190** |          | −0.119*  |          |          | −0.520*** | −0.422*** | −0.480*** | −0.410*** |
| RBS <sup>2</sup>             |          |          | −0.154** |          | −0.139*  |          |          | −0.105*   | −0.061    | −0.053    | −0.088    |
| <b>Mediator</b>              |          |          |          |          |          |          |          |           |           |           |           |
| OR                           |          |          |          | 0.239*** | 0.138*   |          |          |           |           |           |           |
| <b>Moderator</b>             |          |          |          |          |          |          |          |           |           |           |           |
| BDAC                         |          |          |          |          |          |          |          |           | −0.446*** | −0.757*** | −0.514*** |
| <b>Interaction</b>           |          |          |          |          |          |          |          |           |           |           |           |
| PBS*BDAC                     |          |          |          |          |          |          |          |           |           | −0.077    |           |
| PBS <sup>2</sup> *BDAC       |          |          |          |          |          |          |          |           |           | 0.394***  |           |
| RBS*BDAC                     |          |          |          |          |          |          |          |           |           |           | −0.015    |
| RBS <sup>2</sup> *BDAC       |          |          |          |          |          |          |          |           |           |           | 0.114*    |
| R <sup>2</sup>               | 0.127    | 0.163    | 0.214    | 0.175    | 0.227    | 0.156    | 0.174    | 0.352     | 0.523     | 0.560     | 0.530     |
| Adjusted R <sup>2</sup>      | 0.109*** | 0.140*** | 0.186*** | 0.155*** | 0.196*** | 0.138*** | 0.151*   | 0.329***  | 0.504***  | 0.540***  | 0.518*    |
| F                            | 6.954*** | 6.935*** | 7.690*** | 8.665*** | 7.486*** | 8.798*** | 7.490*** | 15.315*** | 27.966*** | 27.359*** | 26.202*** |
| VIF Max.                     | 3.206    | 3.928    | 4.175    | 3.413    | 4.262    | 3.206    | 3.928    | 4.175     | 3.631     | 4.207     | 4.145     |

Note: n = 293, \*p < 0.05, \*\*p < 0.01, \*\*\*p < 0.001.

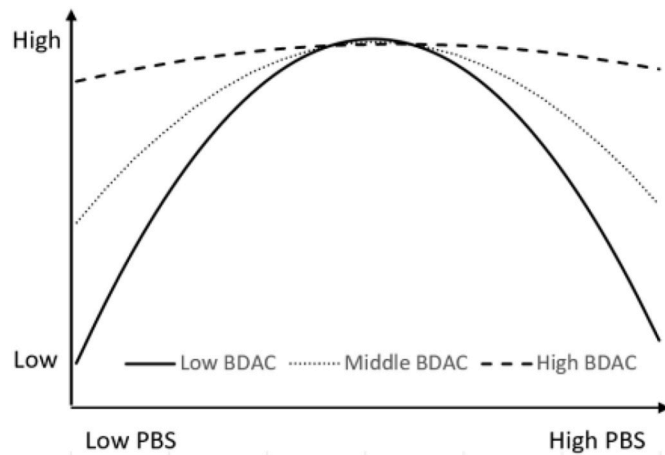


Fig. 2. Moderating role of BDAC between PBS and organizational resilience.

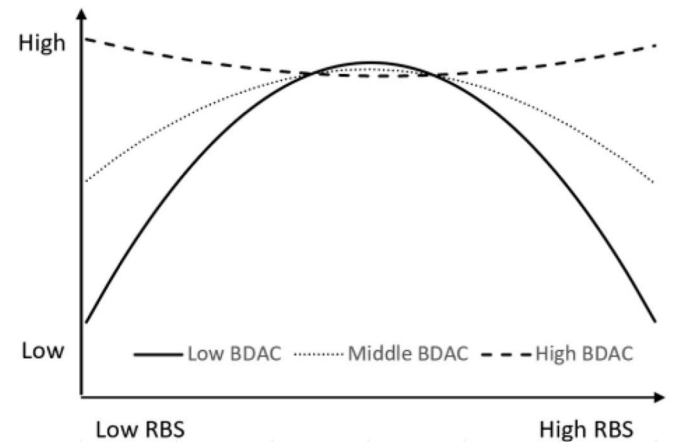


Fig. 3. Moderating role of BDAC between RBS and organizational resilience.

the quadratic term coefficient for RBS decreases and maintains significance ( $\beta = -0.139$ ,  $p < 0.05$ ). This finding further supports hypotheses H4a and H4b.

In Model X and Model XI, BDAC significantly moderates the inverted U-shaped relationship between PBS ( $\beta = 0.657$ ,  $p < 0.001$ ) and RBS ( $\beta = 0.261$ ,  $p < 0.05$ ) with regard to organizational resilience. This finding also supports hypotheses H5a and H5b.

## 5. Conclusions

### 5.1. Discussions

This research tests, based on resource-based theory and dynamic capabilities theory, the nonlinear relationship between boundary-spanning search and organizational resilience and open innovation in the VUCA era, and the moderating role of big data analytic capabilities. Drawing from the analysis of survey data collected from Chinese smart

manufacturing firms, the following research conclusions are obtained.

Firstly, both prospective boundary-spanning search and responsive boundary-spanning search exhibit inverted U-shaped effects on open innovation. The results of the empirical analysis show that the coefficient of the squared term of prospective boundary-spanning search is  $-0.190$ , reaching significance at the 0.01 level. The slope of the curve at its lowest point is 1.315, and at its highest point is  $-0.870$ . The inflection point value is 0.950, which lies within the range of the centered value. The coefficient of the squared term of responsive boundary-spanning search is  $-0.154$ , reaching significance at the 0.01 level. The slope of the curve at its lowest point is 0.909, and at its highest point is  $-0.785$ . The inflection point value is  $-0.019$ , which is within the range of the centered value. Therefore, as the levels of prospective boundary-spanning search and responsive boundary-spanning search increase, open innovation shows a curvilinear change trend of first rising and then falling. Prospective boundary-spanning search enables firms to acquire heterogeneous resources ahead of competitors, conferring a first-mover

**Table 9**  
Results of composite reliability and validity analysis (CLC Estimator tool).

| Variables                                  | Items | Factor loadings | Cronbach's $\alpha$ | CR    | AVE   |
|--------------------------------------------|-------|-----------------|---------------------|-------|-------|
| Prospective Boundary-spanning Search (PBS) | PBS1  | 0.895           | 0.940               | 0.940 | 0.810 |
|                                            | PBS2  | 0.892           |                     |       |       |
|                                            | PBS3  | 0.902           |                     |       |       |
|                                            | PBS4  | 0.900           |                     |       |       |
| Responsive Boundary-spanning Search (RBS)  | RBS1  | 0.790           | 0.900               | 0.900 | 0.690 |
|                                            | RBS2  | 0.838           |                     |       |       |
|                                            | RBS3  | 0.840           |                     |       |       |
|                                            | RBS4  | 0.847           |                     |       |       |
| Organizational Resilience (OR)             | OR1   | 0.867           | 0.950               | 0.950 | 0.770 |
|                                            | OR2   | 0.874           |                     |       |       |
|                                            | OR3   | 0.887           |                     |       |       |
|                                            | OR4   | 0.862           |                     |       |       |
|                                            | OR5   | 0.892           |                     |       |       |
|                                            | OR6   | 0.889           |                     |       |       |
| Big data analytics capabilities (BDAC)     | BDAC1 | 0.894           | 0.970               | 0.970 | 0.810 |
|                                            | BDAC2 | 0.869           |                     |       |       |
|                                            | BDAC3 | 0.907           |                     |       |       |
|                                            | BDAC4 | 0.902           |                     |       |       |
|                                            | BDAC5 | 0.908           |                     |       |       |
|                                            | BDAC6 | 0.909           |                     |       |       |
|                                            | BDAC7 | 0.900           |                     |       |       |
|                                            | BDAC8 | 0.918           |                     |       |       |
| Open Innovation (OI)                       | OI1   | 0.924           | 0.930               | 0.930 | 0.780 |
|                                            | OI2   | 0.910           |                     |       |       |
|                                            | OI3   | 0.834           |                     |       |       |
|                                            | OI4   | 0.865           |                     |       |       |

advantage and furnishing abundant innovative materials for open innovation. However, excessive pursuit of such search may result in information overload, making it difficult to filter and integrate genuinely valuable knowledge amidst a deluge of complex information, ultimately leading to a squander of innovation resources and impeding the progress of open innovation. On the other hand, responsive boundary-spanning search facilitates the acquisition of knowledge that aligns more closely with a firm's existing knowledge base. These fresh knowledge inputs and resource reinforcements foster the expansion of internal innovative thinking and the advancement of open innovation

**Table 10**  
Regression results of hierarchical model (CLC Estimator tool).

|                               | OI       |          |          |           |          | OR       |          |           |           |           |           |
|-------------------------------|----------|----------|----------|-----------|----------|----------|----------|-----------|-----------|-----------|-----------|
|                               | Model I  | ModelII  | ModelIII | ModelIV   | ModelV   | ModelVI  | ModelVII | ModelVIII | ModelIX   | ModelX    | ModelXI   |
| <b>Control Variable</b>       |          |          |          |           |          |          |          |           |           |           |           |
| Industry 2                    | -0.102   | 0.073    | -0.039   | -0.128    | -0.035   | 0.109    | 0.227*** | -0.032    | -0.014    | -0.064    | -0.008    |
| Industry 3                    | 0.015    | 0.049    | -0.037   | -0.055    | -0.056   | 0.291*** | 0.315*** | 0.139     | 0.091     | 0.099     | 0.096     |
| Industry 4                    | 0.140    | 0.308**  | 0.088    | 0.061     | 0.096    | 0.331*** | 0.443*** | -0.055    | -0.015    | -0.070    | -0.002    |
| Industry 5                    | -0.245** | 0.105    | -0.001   | -0.235**  | 0.005    | -0.045   | 0.192    | -0.025    | -0.014    | -0.053    | 0.007     |
| Firm sales                    | -0.172** | -0.133*  | -0.125*  | -0.161**  | -0.124*  | -0.044   | -0.018   | -0.005    | -0.006    | -0.022    | -0.003    |
| Firm age                      | 0.056    | 0.036    | 0.031    | 0.045     | 0.027    | 0.045    | 0.032    | 0.032     | 0.023     | 0.048     | 0.024     |
| <b>Independent variables</b>  |          |          |          |           |          |          |          |           |           |           |           |
| PBS                           |          | 0.283*** | 0.361*** |           | 0.313*** |          | 0.182**  | 0.350***  | 0.236***  | 0.420***  | 0.229**   |
| RBS                           |          | 0.068*   | -0.005   |           | -0.002   |          | 0.069*   | -0.017    | -0.037    | -0.047    | -0.023    |
| <b>Squared</b>                |          |          |          |           |          |          |          |           |           |           |           |
| PBS <sup>2</sup>              |          |          | -0.190** |           | -0.118*  |          |          | -0.519*** | -0.422*** | -0.950*** | -0.409*** |
| RBS <sup>2</sup>              |          |          | -0.155** |           | -0.139*  |          |          | -0.105*   | -0.061    | -0.054    | -0.287*** |
| <b>Mediator</b>               |          |          |          |           |          |          |          |           |           |           |           |
| OR                            |          |          |          | 0.239***  | 0.139*   |          |          |           |           |           |           |
| <b>Moderator</b>              |          |          |          |           |          |          |          |           |           |           |           |
| BDAC                          |          |          |          |           |          |          |          |           | -0.446*** | -0.592*** | -0.465*** |
| <b>Interaction</b>            |          |          |          |           |          |          |          |           |           |           |           |
| PBS*BDAC                      |          |          |          |           |          |          |          |           |           | -0.228*** |           |
| PBS <sup>2</sup> *BDAC        |          |          |          |           |          |          |          |           |           | 0.657***  |           |
| RBS*BDAC                      |          |          |          |           |          |          |          |           |           |           | -0.059    |
| RBS <sup>2</sup> *BDAC        |          |          |          |           |          |          |          |           |           |           | 0.261*    |
| <b>R<sup>2</sup></b>          | 0.129    | 0.165    | 0.216    | 0.177     | 0.228    | 0.156    | 0.174    | 0.352     | 0.522     | 0.560     | 0.529     |
| <b>Adjusted R<sup>2</sup></b> | 0.110*** | 0.141*** | 0.188*** | 0.157***  | 0.198*** | 0.138*** | 0.151*   | 0.329***  | 0.503***  | 0.539***  | 0.508*    |
| <b>F</b>                      | 7.036*** | 7.015*** | 7.763*** | 8.6758*** | 7.558*** | 8.803*** | 7.495*** | 15.299*** | 27.900*** | 27.293*** | 24.149*** |

Note: n = 293, \*p < 0.05, \*\*p < 0.01, \*\*\*p < 0.001.

activities. Nevertheless, overindulgence in responsive boundary-spanning search can trigger knowledge overload within the firm, impeding the effective integration and digestion of excessive external information and resources, thereby diminishing the marginal utility of knowledge and increasing the complexity of open innovation.

Secondly, both prospective boundary-spanning search and responsive boundary-spanning search exert an inverted U-shaped impact on organizational resilience. The results of the empirical analysis show that the coefficient of the squared term of prospective boundary-spanning search is -0.520, reaching significance at the 0.001 level. The slope of the curve at its lowest point is 2.960, and at its highest point is -3.020. The inflection point value is 0.336, which lies within the range of the centered value. The coefficient of the squared term of responsive boundary-spanning search is -0.105, reaching significance at the 0.05 level. The slope of the curve at its lowest point is 0.607, and at its highest point is -0.548. The inflection point value is -0.081, which is within the range of the centered value. Therefore, as the levels of prospective boundary-spanning search and responsive boundary-spanning search increase, organizational resilience shows a curvilinear change trend of first rising and then falling. Acquiring heterogeneous knowledge ahead of competitors enables firms to broaden their knowledge base and respond swiftly to external environmental turbulence and shocks. Nevertheless, excessive engagement in prospective boundary-spanning search consumes considerable time and energy, while inexperienced search results in unnecessary resource wastage, hampering the firm's capacity for rapid adaptation and recovery. Similarly, a moderate level of responsive boundary-spanning search allows organizations to tap into the successes of their partners, enhancing their resilience against external risks. However, over-dependence on responsive boundary-spanning search can deprive firms of the opportunity to proactively adapt and optimize, leading to redundant knowledge accumulation and reducing firms' sensitivity to external threats.

Thirdly, organizational resilience has a significant positive impact on open innovation. The results of the empirical analysis show that the regression coefficient between organizational resilience and open innovation is 0.239, which is significant at the 0.001 level. Therefore, the level of open innovation increases in tandem with the enhancement



of organizational resilience. Organizational resilience enables firms to operate in a stable manner, adapt to changes, absorb new technological knowledge, increase sensitivity to external innovation opportunities and promote open innovation. Additionally, organizational resilience facilitates firms to integrate external resources and forge external connections, thereby enhancing the efficacy of open innovation. Furthermore, organizational resilience enables firms to rapidly recover, withstand risks, maintain a positive attitude towards innovation, dismantle traditional barriers and promote open innovation.

Fourth, organizational resilience partly mediates the influence of boundary-spanning search on open innovation. The results of the empirical analysis show that for the relationship among prospective boundary-spanning search, organizational resilience, and open innovation, when organizational resilience is at low, medium, and high levels, the instantaneous mediating effects are 0.296, 0.061, and  $-0.174$  respectively, and the 95 % confidence interval does not include 0. Therefore, prospective boundary-spanning search not only has a direct effect on open innovation, but also can indirectly influence open innovation through organizational resilience. For the relationship among responsive boundary-spanning search, organizational resilience, and open innovation, when organizational resilience is at lower, medium, and higher levels, the instantaneous mediating effects are 0.175, 0.035, and  $-0.106$  respectively, and the 95 % confidence intervals at the lower and higher levels do not include 0. Therefore, responsive boundary-spanning search not only has a direct effect on open innovation, but also can indirectly influence open innovation through organizational resilience. A moderate level of prospective boundary-spanning search facilitates firms' access to external innovation resources, enhancing their sensitivity and predictive capabilities towards the external environment. This enables firms to effectively identify potential collaborators and jointly promote open innovation. Conversely, a moderate level of responsive boundary-spanning search focuses on complementing and refining existing knowledge, assisting firms in adapting to external environmental changes, promptly adjusting product solutions and processes, and achieving open innovation through the acquisition of partner expertise.

Fifth, big data analytics capability serves as a moderating factor in the inverted U-shaped relationship between prospective and responsive boundary-spanning search and organizational resilience. The results of the empirical analysis show that the coefficient of the squared term of prospective boundary-spanning search in the interaction term with big data analytics capability is 0.394, reaching significance at the 0.001 level. Therefore, as big data analytics capability increases, the curve of prospective boundary-spanning search and open innovation becomes flatter. The coefficient of the interaction term between the squared term of responsive boundary-spanning search and big data analytics capability is 0.114, reaching significance at the 0.05 level. Consequently, as big data analytics capability increases, the curve of responsive boundary-spanning search and open innovation becomes flatter. On one hand, firms that possess high big data analytics capability enhance their ability to integrate and leverage external resources through deep data mining, thereby fostering organizational resilience. Conversely, a low level of big data analytics capability hinders the effective extraction of knowledge gained from prospective boundary-spanning search, resulting in a longer knowledge learning curve and ultimately affecting organizational resilience. On the other hand, a high level of big data analytics capability enables firms to accurately identify boundary-spanning search targets, avoid wasteful search, and efficiently allocate resources. This mitigates the negative impact of excessive search on organizational resilience. Conversely, a low level of big data analytics capability hinders firms' ability to respond promptly to market changes, limiting their capacity to effectively extract and apply knowledge from external resources, thereby weakening organizational resilience.

## 5.2. Theoretical implications

Based on resource-based theory, the nonlinear relationships between prospective and responsive boundary-spanning search and open innovation are clarified. Our findings reveal that both prospective and responsive boundary-spanning searches exhibit an inverted U-shaped influence on open innovation. In other words, a moderate level of both types of searches is most beneficial for firms to successfully engage in open innovation. This discovery aligns with previous studies by Yang et al. (2021) and Zhu et al. (2023), who also observed that the relationship between firms' boundary-spanning search and innovation is not linear. Our study elucidates the dualistic nature of prospective boundary-spanning search and responsive boundary-spanning search as pivotal antecedents of open innovation. Furthermore, it sheds light on the intrinsic motivational mechanism underlying boundary-spanning search and its impact on open innovation.

Utilizing dynamic capabilities theory, we clarify the nonlinear effects of both prospective and responsive boundary-spanning searches on organizational resilience. Our research demonstrates that both types of searches exhibit an inverted U-shaped influence on organizational resilience. Specifically, a moderate level of both searches is most conducive for firms to build and maintain organizational resilience. This finding is consistent with Wang et al. (2023), who similarly concluded that prospective boundary-spanning search has an inverted U-shaped effect on both prospective and improvisational resilience. Furthermore, our study reveals that both types of boundary-spanning searches play a significant role in enhancing organizational resilience, thereby expanding the current understanding of organizational resilience.

Based on dynamic capabilities theory, the mediating role of organizational resilience in the relationship between prospective and responsive boundary-spanning search and open innovation is elucidated. Our findings reveal that organizational resilience partially mediates the association between both prospective and responsive boundary-spanning searches and open innovation. This discovery aligns with the research conducted by Wang et al. (2023), who determined that organizational resilience serves as a mediator between prospective boundary-spanning search and green innovation. Our study further delves into the formation mechanism of open innovation, validates the crucial role of organizational resilience in fostering open innovation, and augments the existing research on the antecedents and consequences of organizational resilience.

Drawing on dynamic capabilities theory, the moderating influence of big data analytics capability on the nonlinear relationship between boundary-spanning search and organizational resilience is clarified. Our research demonstrates that high big data analytics capability flattens the inverted U-shaped relationship between both prospective and responsive boundary-spanning searches and organizational resilience. This study illuminates the moderating effect of big data analytics capability on organizational resilience, offering deeper insights into the role of prospective and responsive boundary-spanning search in enhancing organizational resilience at different levels of big data analytics capability. The study reveals the intrinsic mechanism of prospective and responsive boundary-spanning search on open innovation through organizational resilience and big data analytical capability in the digital era, and provides theoretical support for smart manufacturing firms to carry out open innovation.

## 5.3. Managerial implications

First, firms must strike a balance when conducting boundary-spanning search. As the saying goes, "too much is as bad as too little." The scope of prospective and responsive boundary-spanning searches should be limited to a reasonable range in order to optimize their beneficial impacts and mitigate potential adverse effects. Managers should distribute resources between prospective and responsive boundary-spanning searches based on the firm's stage of development

and strategic objectives. For start-ups or firms pursuing breakthrough innovations, increasing investments in prospective boundary-spanning search can help explore new technologies, markets, and partnerships, thereby creating opportunities for new development avenues. For mature firms, balancing prospective and responsive searches is essential. Managers should ensure a certain degree of foresight when increasing responsive search to address market demands effectively while quickly adjusting resource allocation to meet changing conditions. Additionally, those responsible for managing smart manufacturing companies must maintain a balance between prospective and responsive boundary-spanning searches. Regularly evaluating the efficiency and effectiveness of these searches can dynamically adjust search strategies in response to market changes or competitive pressures. Prospective boundary-spanning search enables firms to identify emerging opportunities and trends, while responsive boundary-spanning search allows firms to respond to immediate threats and market changes. Excessive reliance on either type of search may result in missed opportunities or delayed responses. Therefore, managers should develop a unified strategy combining prospective and responsive boundary-spanning searches to achieve a balance between innovation efficiency and the technological advantages of featured products. A comprehensive boundary-spanning search program ensures that smart manufacturing firms are fully informed about external environmental changes and market dynamics. Conversely, well-developed search system can enhance access to knowledge and information while allowing adjustments in search priorities based on real-time market feedback or changes.

Second, boundary-spanning search play a crucial role in enhancing organizational resilience by promoting open innovation, which enables firms to adapt effectively to dynamic market conditions. Organizational resilience refers to the ability of an organization to quickly recover from internal and external challenges while maintaining or improving its competitive position. Prospective boundary-spanning search involves exploring new technologies, emerging markets, and potential partnership avenues beyond current boundaries, helping firms identify opportunities and threats in the changing environment. By integrating these discoveries with existing resources through targeted strategies, firms can build organizational resilience by adapting to external challenges. At the same time, prospective boundary-spanning search aids in adjusting internal strategies to better align with market dynamics. Responsive boundary-spanning search complements this process by enabling firms to monitor competitors' successes and failures in technological innovation and market strategies, allowing them to respond swiftly without deterring internal development. By learning from best practices of industry leaders, firms optimize their resource allocation and operational processes, enhancing efficiency and flexibility. This includes improving supply chain management or production optimization to reduce costs and improve operational efficiency. Together, prospective boundary-spanning search and responsive boundary-spanning search help firms continuously integrate, build, and reconfigure resources through various mechanisms, enabling them to perceive, respond to, and adapt to environmental changes effectively. In doing so, firms can enhance organizational resilience, which is critical for maintaining competitive advantages and achieving sustainable growth in complex and volatile markets. To proactively build organizational resilience, managers should take steps such as providing systematic training for employees, covering skills like interdisciplinary teamwork, rapid learning, and adaptation to change. Additionally, optimizing the organizational structure and establishing a flexible operating mechanism can facilitate the rapid adjustment of resource allocation when market conditions change. In this way, developing flexible contingency plans can ensure a firm's capacity to maintain stable operations in an uncertain environment, thereby promoting open innovation.

Finally, in the digital age, smart manufacturing firms must leverage big data technologies to enhance decision-making, resource allocation, and real-time monitoring of boundary-spanning search activities. By

analyzing market trends, customer demand, and internal firms' data comprehensively, managers can better assess the effectiveness of different search strategies and adjust their focus accordingly. For instance, analyzing market trends and customer feedback can provide a deeper understanding of market dynamics, enabling firms to develop more targeted and effective management strategies. Smart manufacturing firms need to approach the implementation of big data analytics capabilities in several ways. For instance, they must establish a comprehensive data management system that includes data collection, storage, organization, and analysis, ensuring data quality and security. Moreover, they should introduce advanced big data analysis technologies and tools, such as machine learning and artificial intelligence, to improve efficiency and accuracy in data processing. In addition, they need to cultivate professional big data analysis talents and establish data analysis teams to provide the necessary support for implementing these capabilities. By integrating these approaches, smart manufacturing firms can harness big data analytics to drive innovation, optimize operations, and achieve sustainable growth in the digital age.

In particular, Haier, a prominent entity within the home appliance sector, not only persists in pioneering advancements within the domain of home appliances but also diversifies its operations to encompass the field of smart homes. Firstly, Haier acquires cutting-edge technologies such as the Internet of Things and big data through prospective boundary-spanning searches, applies these technologies to home appliances through independent research and development or collaboration with Tencent Cloud and Huaxida, and jointly develops smart home solutions to realize open innovation. Concurrently, Haier is also proactively developing an industrial Internet platform to achieve the transformation from a home appliance manufacturer to an Internet of Things eco-brand. Secondly, Haier has conducted responsive boundary-spanning search in the field of smart home appliances and launched a series of innovative products, including smart refrigerators and smart washing machines. These products not only satisfy consumers' demand for smart home appliances, but also facilitate the intelligent development of the home appliance industry. Consequently, Haier has consistently expanded its business scope and market reach through prospective and responsive boundary-spanning search, thereby enhancing the firm's risk resilience and market competitiveness. This resilience enables Haier to promptly adjust its strategic orientation to response to shifts in the external environment, and provide uninterrupted and accurate services to users throughout the entire process of demand identification, order channels, marketing and sales, order management, service operation and maintenance, and user segmentation. This rapid response capability enables Haier to maintain a leading position in fierce market competition.

#### 5.4. Limitations and future research

Firstly, our study employs cross-sectional data to investigate the intricate relationship between prospective and responsive boundary-spanning search and open innovation. However, future research can use longitudinal data and experimental methods to enhance causal reasoning, and delve deeper into whether the impact of prospective and responsive boundary-spanning search and open innovation varies at different stages of a firm's lifecycle. This approach will offer a more nuanced understanding of how these factors evolve and interact over time.

Secondly, while we emphasized the significance of big data analytics capability in explaining the relationship between prospective and responsive boundary-spanning search, organizational resilience, and open innovation, our analysis did not delve into its disaggregated dimensions. Future studies could categorize big data analytics capability into two primary components: big data technical capabilities and big data application capabilities. Such an analysis would provide a more comprehensive understanding of how big data analytics capability influence organizational open innovation. Furthermore, the moderating

influence of internal factors (including organizational cultural and leadership styles) and external environmental factors (such as regulatory environment, competitive intensity and technological maturity) on the relationship between prospective and responsive boundary-spanning search, organizational resilience, and open innovation provide a promising avenue for future research. Finally, future research can focus on analyzing this phenomenon in different geographical or industry contexts.

### CRedit authorship contribution statement

**Shumo Jin:** Writing – original draft, Formal analysis, Conceptualization. **Juanru Wang:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition. **Peiyu Zhu:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation. **Robert Moehler:** Writing – review & editing, Software, Resources.

### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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### Data availability

Data will be made available on request.

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